

DBMS PROJECT PRESENTATION

SPORTS CLUB MEMBERSHIP AND TOURNAMENT MANAGEMENT SYSTEM

DATABASE MANAGEMENT SYSTEM PROJECT

STUDENT NAME

Sheikh Arsh Ali

ROLL NUMBER

25WU0102255

SUBJECT

DBMS

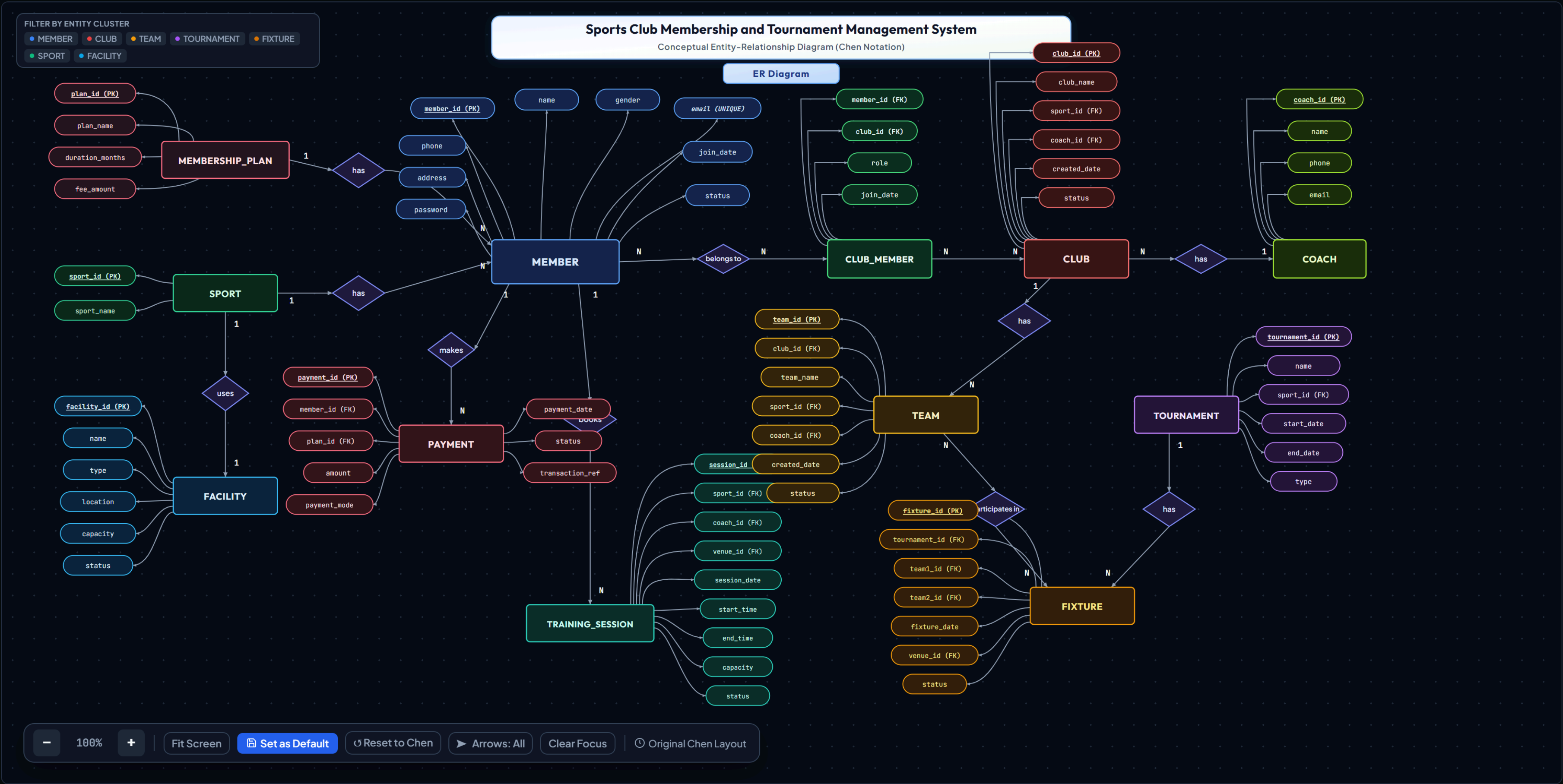
FACULTY NAME

Dr. Kiran Mayee

CONCEPTUAL DESIGN: ENTITY RELATIONSHIP DIAGRAM

Interactive conceptual architecture — 12 Entities, 10 Relationships, 69 Attributes, live drag & drop.

[Open in Full Screen / Studio](#)



RELATIONAL DATABASE DESIGN

Schema definitions, constraints, foreign keys, and 3NF normalization

MEMBERSHIP_PLAN			Table 01 • Plan Tier
FIELD	DATA TYPE	CONSTRAINT	
plan_id	INTEGER	PK	
plan_name	VARCHAR(50)		
duration_months	INTEGER		
fee_amount	NUMERIC(10,2)		

MEMBER			Table 02 • Core User
FIELD	DATA TYPE	CONSTRAINT	
member_id	INTEGER	PK	
name	VARCHAR(100)		
gender	VARCHAR(10)		
phone	VARCHAR(20)		
email	VARCHAR(100)	UNIQUE	
password	VARCHAR(255)		
address	VARCHAR(255)		
join_date	DATE		
status	VARCHAR(20)		

SPORT			Table 03 • Discipline
FIELD	DATA TYPE	CONSTRAINT	
sport_id	INTEGER	PK	
sport_name	VARCHAR(50)	UNIQUE	

RELATIONAL DATABASE DESIGN

Schema definitions, constraints, foreign keys, and 3NF normalization

FACILITY			Table 04 • Venue
FIELD	DATA TYPE	CONSTRAINT	
facility_id	INTEGER	PK	
name	VARCHAR(100)		
type	VARCHAR(50)		
location	VARCHAR(100)		
capacity	INTEGER		
status	VARCHAR(20)		

COACH			Table 05 • Instructor
FIELD	DATA TYPE	CONSTRAINT	
coach_id	INTEGER	PK	
name	VARCHAR(100)		
phone	VARCHAR(20)		
email	VARCHAR(100)	UNIQUE	

CLUB			Table 06 • Team
FIELD	DATA TYPE	CONSTRAINT	
club_id	INTEGER	PK	
club_name	VARCHAR(100)		
sport_id	INTEGER	FK	
→ SPORT			
coach_id	INTEGER	FK	
→ COACH			
created_date	DATE		
status	VARCHAR(20)		

RELATIONAL DATABASE DESIGN

Schema definitions, constraints, foreign keys, and 3NF normalization

CLUB_MEMBER			Table 07 • Bridge (N:N)
FIELD	DATA TYPE	CONSTRAINT	
member_id → MEMBER	INTEGER	PK, FK	
club_id → CLUB	INTEGER	PK, FK	
role	VARCHAR(50)		
join_date	DATE		

PAYMENT			Table 08 • Ledger
FIELD	DATA TYPE	CONSTRAINT	
payment_id	INTEGER	PK	
member_id → MEMBER	INTEGER	FK	
plan_id → MEMBERSHIP_PLAN	INTEGER	FK	
amount	NUMERIC(10,2)		
payment_mode	VARCHAR(30)		
payment_date	DATE		
status	VARCHAR(20)		
transaction_ref	VARCHAR(100)	UNIQUE	

TRAINING_SESSION			Table 09 • Clinic
FIELD	DATA TYPE	CONSTRAINT	
session_id	INTEGER	PK	
sport_id → SPORT	INTEGER	FK	
coach_id → COACH	INTEGER	FK	
venue_id → FACILITY	INTEGER	FK	
session_date	DATE		
start_time	TIME		
end_time	TIME		
capacity	INTEGER		
status	VARCHAR(20)		

RELATIONAL DATABASE DESIGN

Schema definitions, constraints, foreign keys, and 3NF normalization

TEAM			Table 10 • Squad
FIELD	DATA TYPE	CONSTRAINT	
team_id	INTEGER	PK	
club_id → CLUB	INTEGER	FK	
team_name	VARCHAR(100)		
sport_id → SPORT	INTEGER	FK	
coach_id → COACH	INTEGER	FK	
created_date	DATE		
status	VARCHAR(20)		

TOURNAMENT			Table 11 • Event
FIELD	DATA TYPE	CONSTRAINT	
tournament_id	INTEGER	PK	
name	VARCHAR(100)		
sport_id → SPORT	INTEGER	FK	
start_date	DATE		
end_date	DATE		
type	VARCHAR(50)		

FIXTURE			Table 12 • Match
FIELD	DATA TYPE	CONSTRAINT	
fixture_id	INTEGER	PK	
tournament_id → TOURNAMENT	INTEGER	FK	
team1_id → TEAM	INTEGER	FK	
team2_id → TEAM	INTEGER	FK	
fixture_date	DATE		
venue_id → FACILITY	INTEGER	FK	
status	VARCHAR(20)		

FILLED-IN RDBMS

Live PostgreSQL Instances • 12 Populated Relational Tables • Atomic Records

FILLED-IN RDBMS DATA INSTANCES

Populated PostgreSQL tables • Configuration, Sports Master & Infrastructure Venues

MEMBERSHIP_PLAN				5 Tiers Populated
PLAN_ID	PLAN_NAME	DURATION	FEE_AMOUNT	
101	Standard Monthly Pass	1 Month	₹499.90	
102	Quarterly Fitness Pass	3 Months	₹1399.90	
103	Annual Gold Elite Pass	12 Months	₹4999.90	
104	Student Semester Pass	6 Months	₹1999.90	
105	Weekend Warrior Pass	1 Month	₹299.90	

SPORT		6 Disciplines Populated
SPORT_ID	SPORT_NAME	
201	Football	
202	Cricket	
203	Badminton	
204	Basketball	
205	Tennis	
206	Swimming	

FACILITY					6 Arenas Populated
ID	NAME	TYPE	CAPACITY	STATUS	
301	Main Arena Stadium Turf	Outdoor Football Pitch	1500	Available	
302	Imperial Cricket Oval	Outdoor Cricket Ground	2500	Booked	
303	Grand Slam Center Court	Clay & Synthetic Court	350	Booked	
304	Smash Point Badminton Arena	Indoor Wooden Court	120	Available	
305	Skyline Basketball Dome	Indoor Hardwood Court	400	Booked	
306	Olympic Aquatic Center	Olympic Pool (50m)	250	Under Maintenance	

FILLED-IN RDBMS DATA INSTANCES

Populated PostgreSQL tables • Instructors, Club Franchises & Competitive Squads

COACH				6 Instructors
ID	NAME	PHONE	EMAIL	
401	Carlos Mendez	+91-9820154321	carlos.mendez@gmail.com	
402	Vikram Singhanian	+91-9811234567	vikram.singh@gmail.com	
403	Elena Rostova	+91-9717890123	elena.rostova@gmail.com	
404	Lee Wei Chen	+91-9876543210	lee.wei@gmail.com	
405	Marcus Aurelius Hayes	+91-9945012345	marcus.hayes@gmail.com	
406	Sarah Jenkins	+91-9848098765	sarah.jenkins@gmail.com	

CLUB					6 Clubs
ID	CLUB_NAME	SPORT	COACH	STATUS	
501	Thunderbolts FC	201 (Football)	401 (C. Mendez)	Active	
502	Apex Strikers CC	202 (Cricket)	402 (V. Singhanian)	Active	
503	Ace Masters TC	205 (Tennis)	403 (E. Rostova)	Active	
504	Smash Point BC	203 (Badminton)	404 (L. W. Chen)	Active	
505	Royal Hoops BC	204 (Basketball)	405 (M. A. Hayes)	Active	
506	Aqua Dolphins SC	206 (Swimming)	406 (S. Jenkins)	Active	

TEAM (SQUADS)				8 Squads
ID	TEAM_NAME	CLUB	STATUS	
701	Thunderbolts Senior A	501 (Thunderbolts FC)	Active	
702	Thunderbolts Under-21	501 (Thunderbolts FC)	Active	
703	Apex Strikers XI	502 (Apex Strikers CC)	Active	
704	Apex Colts XI	502 (Apex Strikers CC)	Active	
705	Ace Masters Pro Squad	503 (Ace Masters TC)	Active	
706	Smash Point Shuttlers	504 (Smash Point BC)	Active	
707	Royal Hoops Varsity	505 (Royal Hoops BC)	Active	
708	Royal Hoops Ballers	505 (Royal Hoops BC)	Active	

FILLED-IN RDBMS DATA INSTANCES

Populated PostgreSQL tables • Registered Member Athletes (27 Members) & 1NF Atomic Roles

MEMBER					27 Registered Athletes (@gmail.com)	
ID	NAME	GENDER	PHONE	EMAIL	ADDRESS	STATUS
1001	Aarav Sharma	Male	+91-9820011223	aarav.sharma@gmail.com	Sector 14, Mumbai	Active
1002	Sophia Martinez	Female	+91-9819922334	sophia.martinez@gmail.com	Bandra West, Mumbai	Active
1003	Daniel Kim	Male	+91-9769933445	daniel.kim@gmail.com	Koramangala, Bengaluru	Expired
1004	Fatima Al-Mansoor	Female	+91-9920044556	fatima.mansoor@gmail.com	Marine Lines, Mumbai	Active
1005	Lucas Silva	Male	+91-9833055667	lucas.silva@gmail.com	Indiranagar, Bengaluru	Active
1006	Priya Nair	Female	+91-9870066778	priya.nair@gmail.com	Silver Oak, Kochi	Pending
1007	Liam O'Connor	Male	+91-9821077889	liam.oconnor@gmail.com	Jubilee Hills, Hyderabad	Active
1008	Chloe Dubois	Female	+91-9987088990	chloe.dubois@gmail.com	Whitefield, Bengaluru	Expired
1009	Rohan Gupta	Male	+91-9810099001	rohan.gupta@gmail.com	HITEC City, Hyderabad	Active
1010	Aisha Hassan	Female	+91-9871100112	aisha.hassan@gmail.com	DLF Phase 5, Gurugram	Active
1011	Mateo Rossi	Male	+91-9818811223	mateo.rossi@gmail.com	Vasant Vihar, New Delhi	Suspended

CLUB_MEMBER (1NF Atomic Roles)			34 Assignments
MEMBER_ID	CLUB_ID	ROLE	JOIN_DATE
1002	501 (Thunderbolts FC)	Captain	2024-01-16
1002	501 (Thunderbolts FC)	Striker	2024-01-16
1005	501 (Thunderbolts FC)	Vice-Captain	2024-02-12
1005	501 (Thunderbolts FC)	Midfielder	2024-02-12
1011	501 (Thunderbolts FC)	Defender	2024-04-13
1016	501 (Thunderbolts FC)	Goalkeeper	2024-06-16
1020	501 (Thunderbolts FC)	Winger	2024-08-21
1027	501 (Thunderbolts FC)	Forward	2024-12-06
1001	502 (Apex Strikers CC)	Captain	2024-01-13
1001	502 (Apex Strikers CC)	All-Rounder	2024-01-13
1007	502 (Apex Strikers CC)	Opening Batsman	2024-03-02
1009	502 (Apex Strikers CC)	Wicket-Keeper	2024-03-16
1018	502 (Apex Strikers CC)	Fast Bowler	2024-07-15
1024	502 (Apex Strikers CC)	Spin Bowler	2024-10-16
1014	503 (Ace Masters TC)	Captain	2024-05-19

FILLED-IN RDBMS DATA INSTANCES

Populated PostgreSQL tables • Transactions Ledger, Coaching Clinics & Tournament Matches

PAYMENT (Ledger)							30 Financial Transactions
ID	MEMBER	PLAN	AMOUNT	MODE	STATUS	TX_REF	
8001	1001	103	₹4999.90	UPI	Paid	TXN_20240112_001	
8002	1002	103	₹4999.90	Credit Card	Paid	TXN_20240115_002	
8003	1003	101	₹499.90	Cash	Expired	TXN_20240120_003	
8004	1004	102	₹1399.90	Net Banking	Paid	TXN_20240202_004	
8005	1005	103	₹4999.90	Credit Card	Paid	TXN_20240210_005	
8006	1006	104	₹1999.90	Bank Transfer	Pending	TXN_20240218_006	
8007	1007	102	₹1399.90	UPI	Paid	TXN_20240301_007	

TRAINING_SESSION							9 Clinics
ID	SPORT	COACH	VENUE	DATE	TIME	STATUS	
601	201 (Football)	401 (C. Mendez)	301 (Main Arena)	2025-02-15	07:00 - 09:00	Completed	
602	202 (Cricket)	402 (V. Singhania)	302 (Imperial Oval)	2025-02-16	08:00 - 11:00	Completed	
603	205 (Tennis)	403 (E. Rostova)	303 (Grand Slam)	2025-02-17	16:00 - 18:00	Completed	
604	203 (Badminton)	404 (L. W. Chen)	304 (Smash Point)	2025-02-18	17:00 - 19:00	Completed	
605	204 (Basketball)	405 (M. A. Hayes)	305 (SkyLine Dome)	2025-02-19	18:00 - 20:00	Completed	
606	206 (Swimming)	406 (S. Jenkins)	306 (Olympic Pool)	2025-02-20	06:30 - 08:30	Completed	
607	201 (Football)	401 (C. Mendez)	301 (Main Arena)	2026-10-05	07:00 - 09:00	Scheduled	

TOURNAMENT					3 Championships
ID	NAME	SPORT	DATES	FORMAT	
9001	Inter-Club Football Super League	201 (Football)	2025-03-01 to 2025-03-20	League	
9002	Champions Cricket Cup 2025	202 (Cricket)	2025-04-05 to 2025-04-25	Round Robin	
9003	Autumn Invitational Basketball Trophy	204 (Basketball)	2025-05-10 to 2025-05-25	Knockout	

FIXTURE (Match Schedule)						6 Squad Matches
ID	TOURNAMENT	TEAM 1 VS TEAM 2	DATE	VENUE	STATUS	
911	9001 (Super League)	701 (Thunderbolts Sr A) vs 702 (Thunderbolts U-21)	2025-03-05 16:00	301 (Main Arena)	Completed	
912	9001 (Super League)	702 (Thunderbolts U-21) vs 701 (Thunderbolts Sr A)	2025-03-12 16:00	301 (Main Arena)	Completed	
913	9002 (Cricket Cup)	703 (Apex Strikers XI) vs 704 (Apex Strikers II)	2025-04-10 10:00	302 (Cricket Oval)	Completed	

SQL QUERY EXECUTION & RESULT SETS